

RESEARCH PROJECT

Data-Driven Early Detection of Thermal Runaway in Lithium-ion Batteries

Shubha Sankar Banerjee^{*1}, Heqiao Ruan^{*1}, Mustapha Makki³, Paul Leu², Satish Iyengar¹

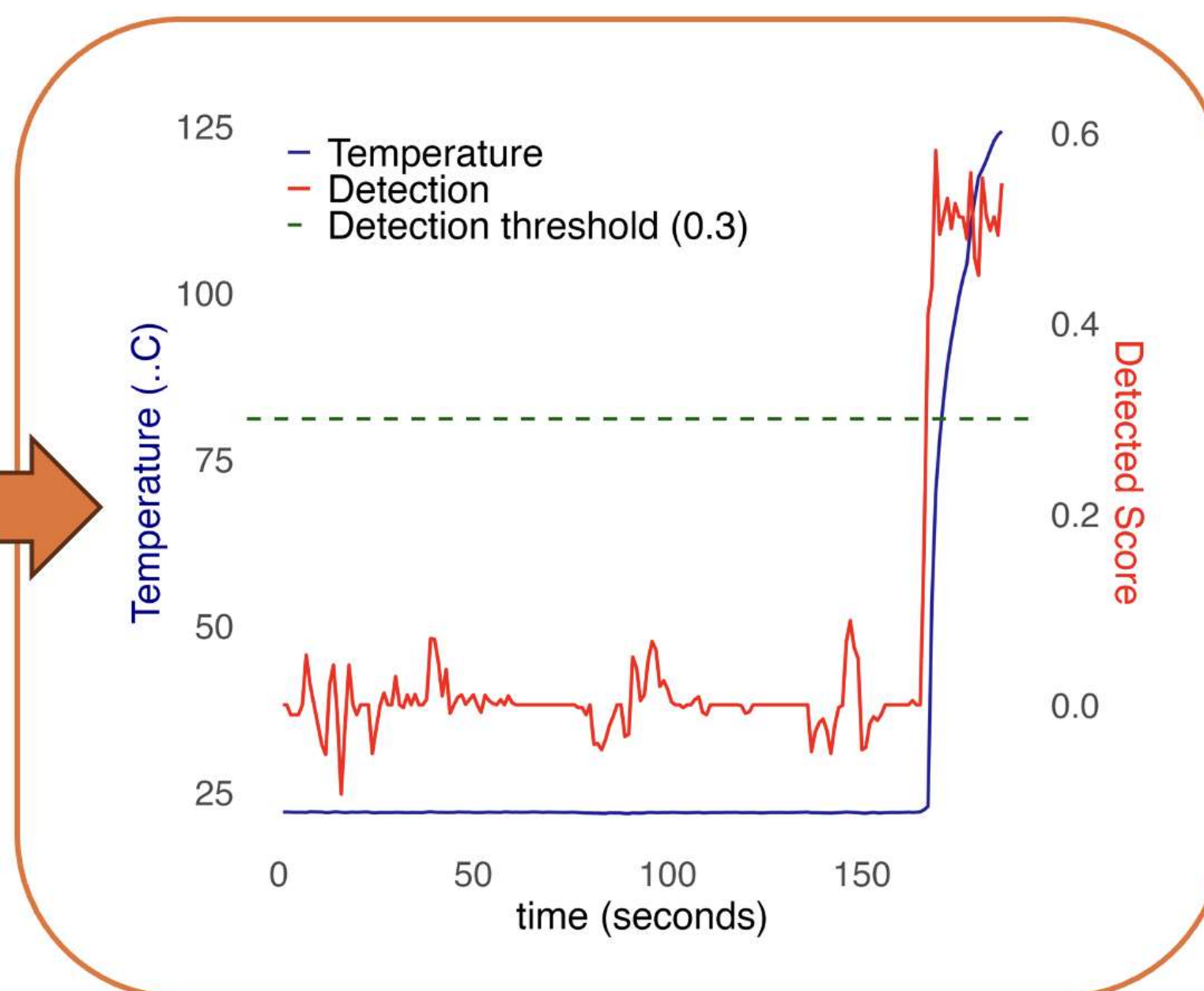
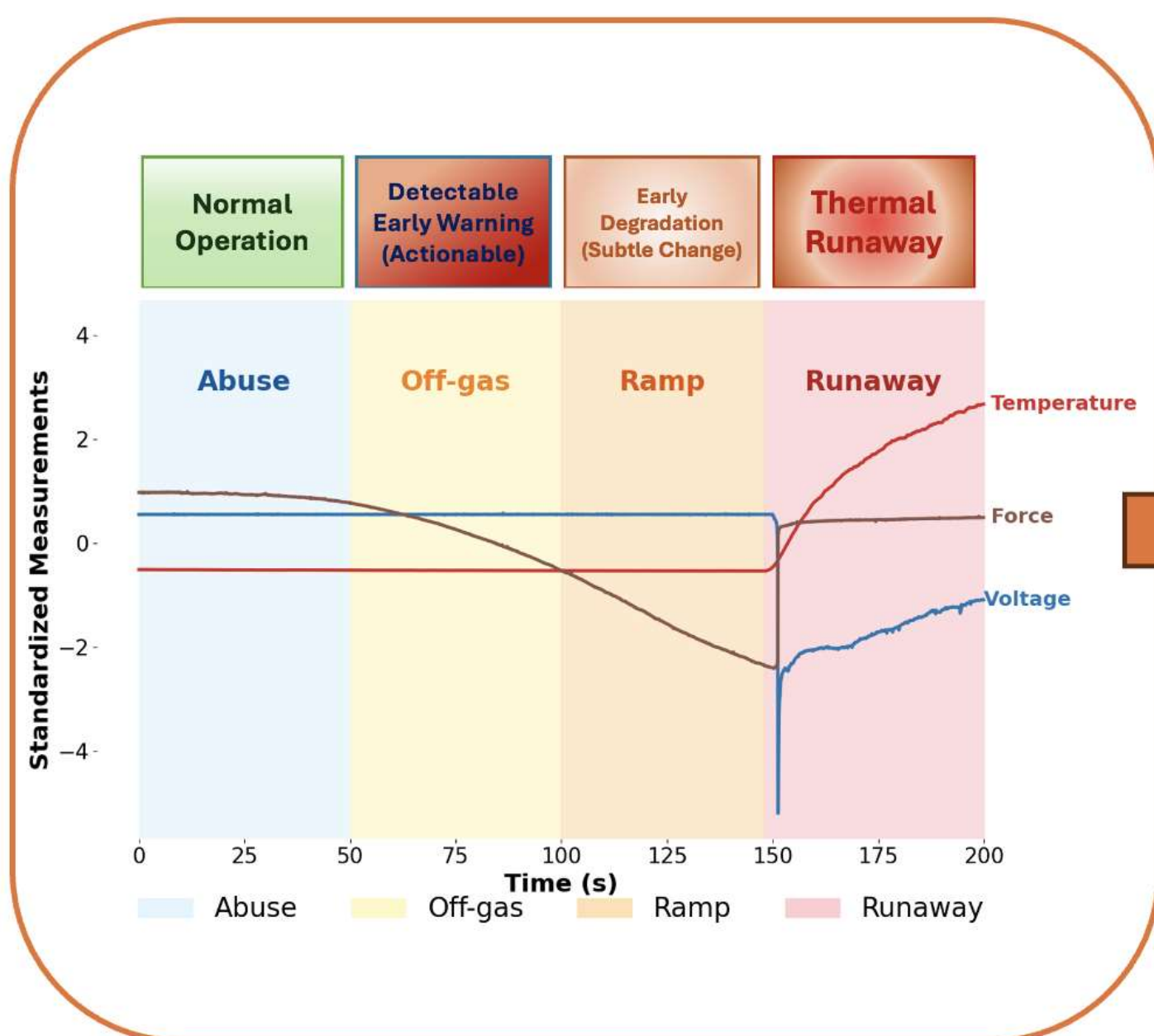
Department of Statistics¹, Department of Industrial Engineering², University of Pittsburgh, Eaton Research Lab³

April 15, 2026

Methodology

Motivation

- Safety Critical:**
Thermal Runaway (TR) poses severe fire/explosion risks.
- Sensor Limits:**
High-end sensors (gas/acoustic) are too costly for mass deployment.
- Data Scarcity:**
TR is rare; black-box models lack enough failure data or interpretability of results.
- Standard Signals:**
Need for detection using only existing Battery Management Systems (BMS) telemetry.
- Detection Delay:**
Thresholds often miss early-stage internal shifts.



Outcomes

- Online Deployment:**
Real-time monitoring for streaming BMS data.
- SVD aggregation:**
Multi-channel fusion via dimensionality reduction.
- Software Ready:**
Functional implementations in R and Python by deploying packages [TRpred](#) and [pTRpred](#) respectively.
- Interpretable Logic:**
Transparent, slope-based statistical flagging
- Broadly Adaptive:**
Flexible & generalizable across different chemistries and scales

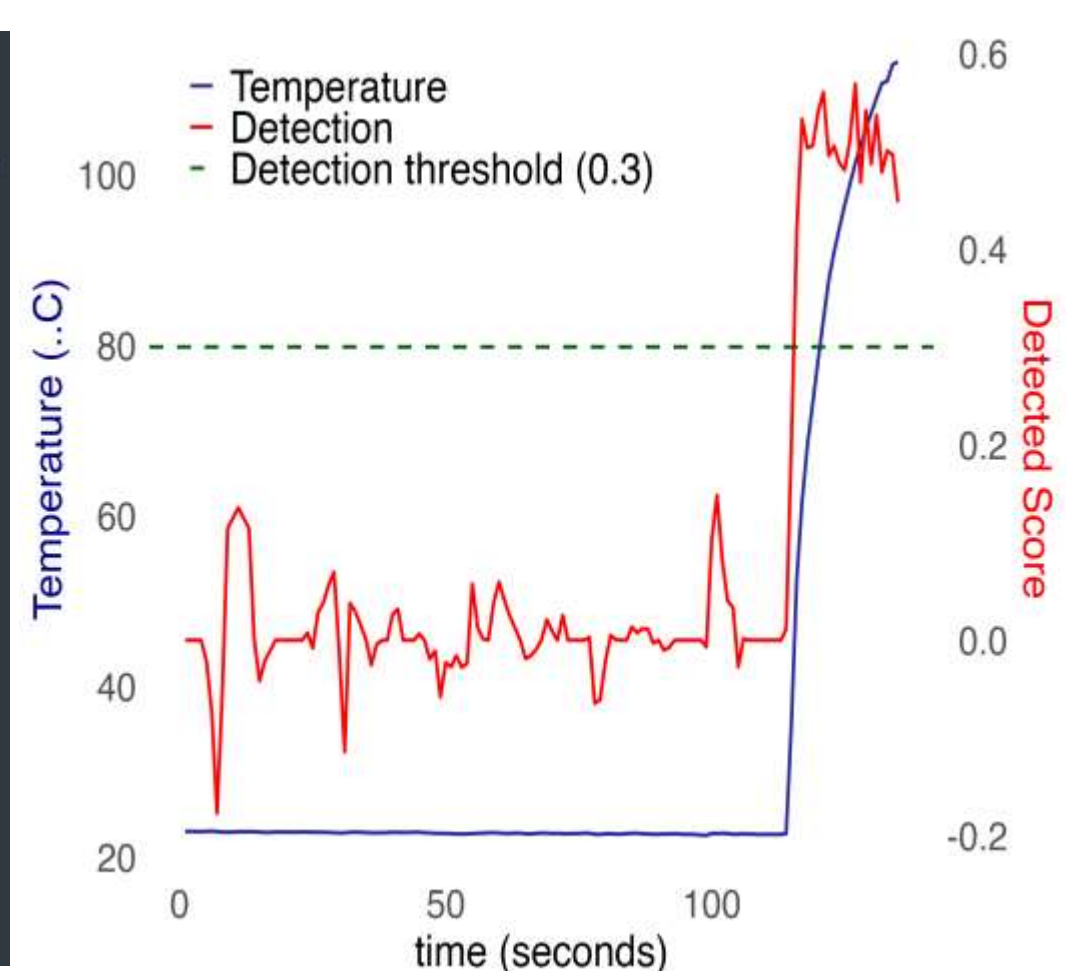
1. Data from Battery: Temperature, SOC, Voltage etc. - Multivariate Time Series
2. Generate the pre-signal thermal runaway ideally in the early degradation stage

1. Specify a running window with varying size to calculate the slope for each of the period.
2. Calculate proportion of periods with abnormally high slope($3 * MAD$) as the real-time detected score

Software Implementation in Python

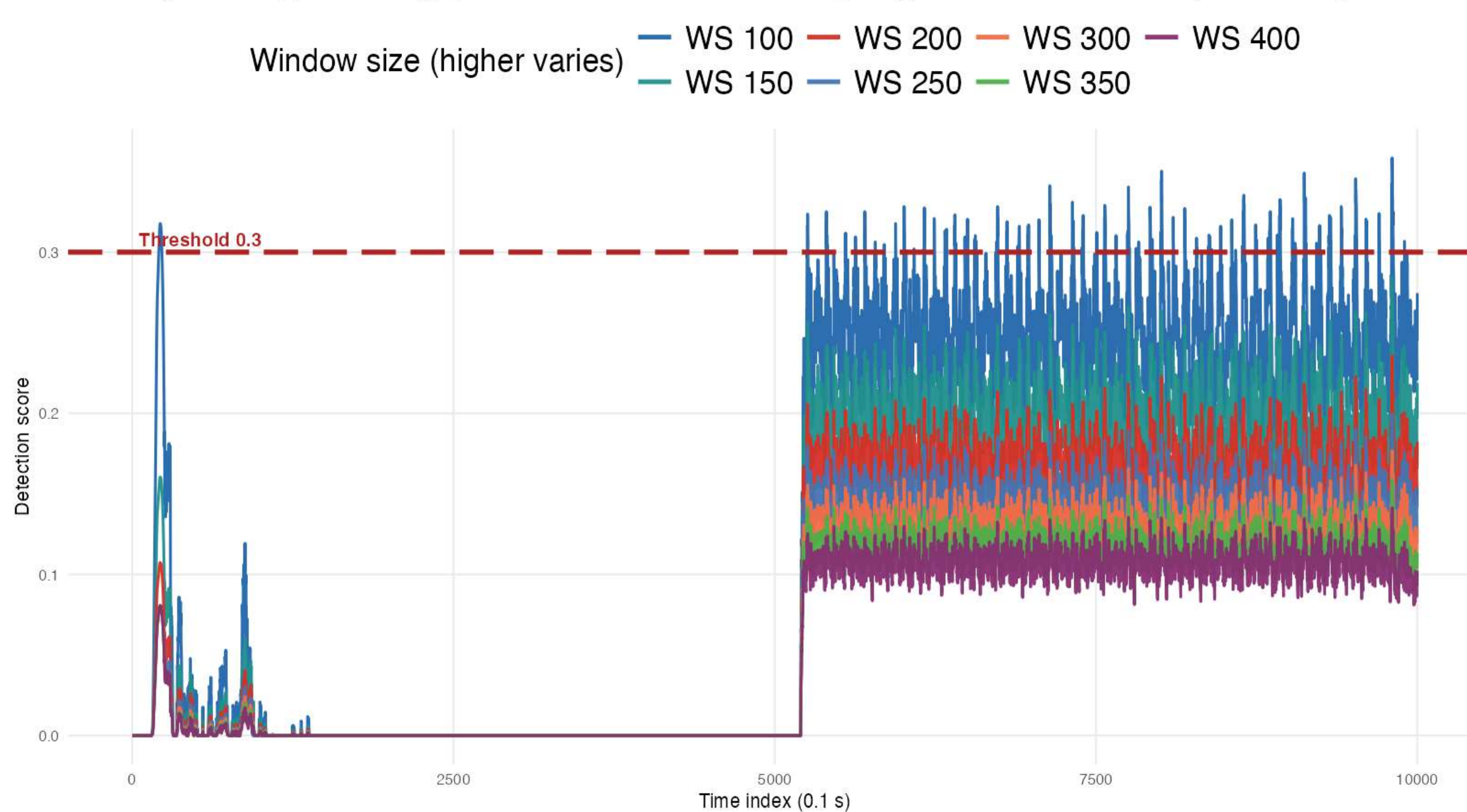
```
def detect_realtime(
    time: Sequence,
    signal: Union[Sequence[float], np.ndarray],
    lowwl: int = 5,
    highwl: Union[str, int] = "auto",
    mad_k: float = 3.0,
    direction: str = "positive",
    burn_in: Optional[int] = None,
    smooth_k: int = 30,
    threshold: float = 1.3,
):
    ...

def detect_asvotes(signal, lowwl=5, highwl="auto", mad_k=3, direction="positive"):
```



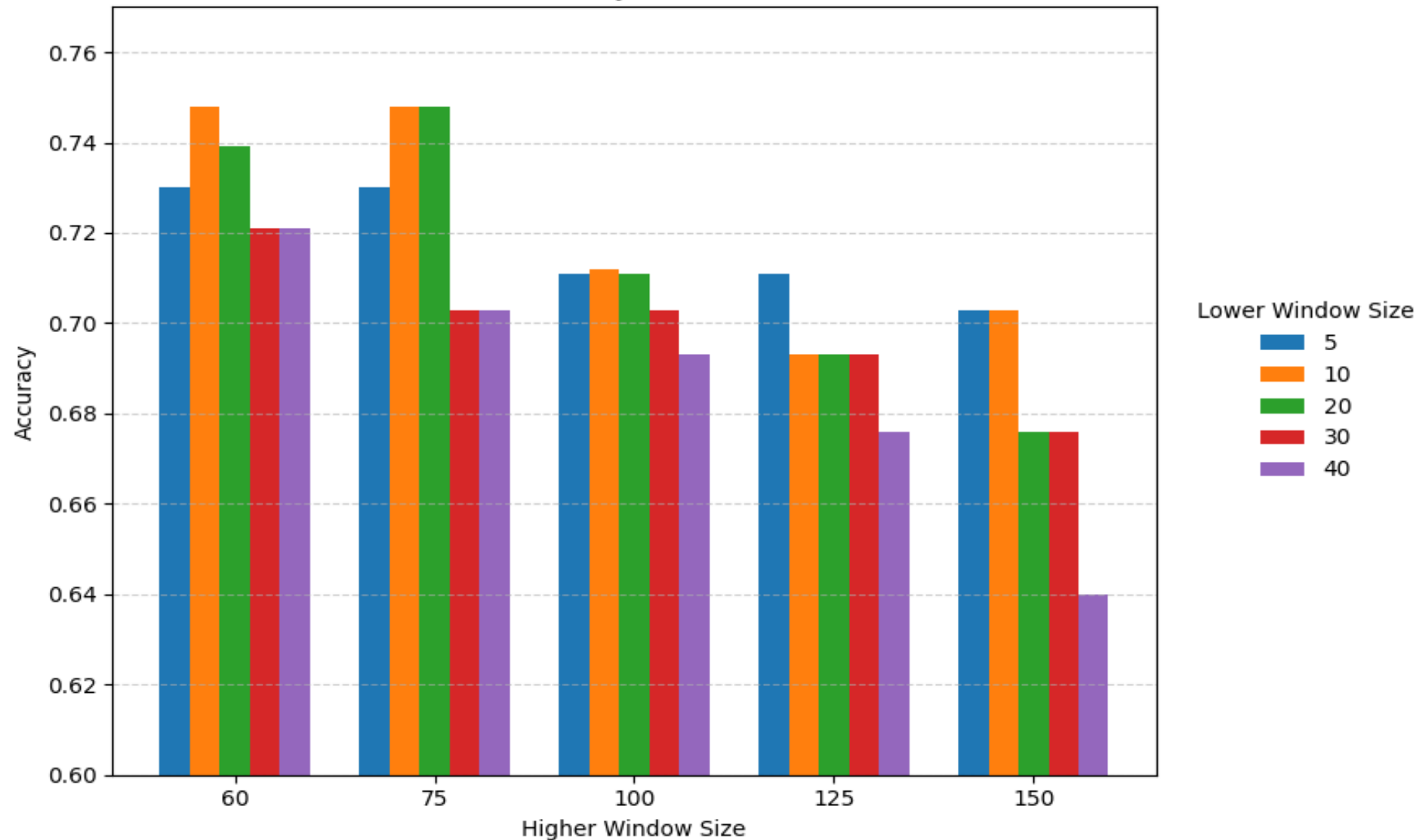
Results on Simulated Data

batteryCCCV_ordinary_54 — Detection score by higher window size (lower 50)

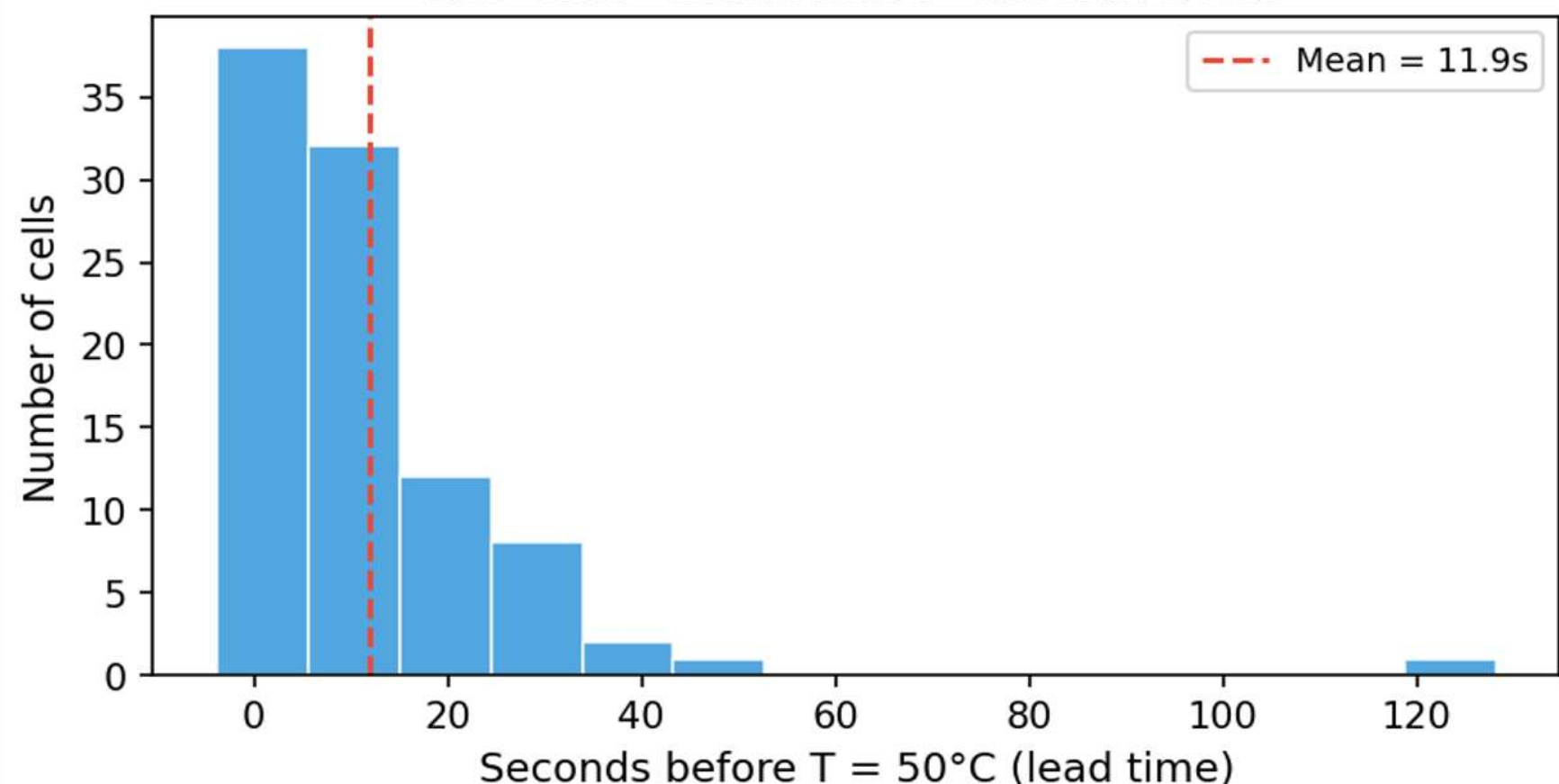


Results for Real Data

SVD-Based Accuracy across Window Sizes



118 cells · mean 11.9s · median 10.1s



Threshold Sensitivity Analysis

